**National University of Computer & Emerging Sciences, Karachi**

**Department of Computer Science**

**SPRING 2022**

Sohaib Sarosh Shamsi

| **Course Code:** EE-1005 | **Course Name:** Digital Logic Design |
|---|---|
| **Course Teacher:** Aamir Ali | **Assignment No:** 03 |

21k-3278

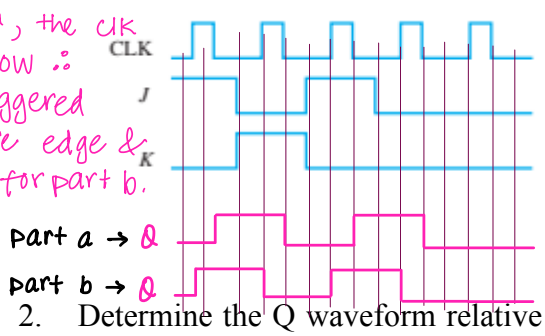
***Instructions for Submission:***

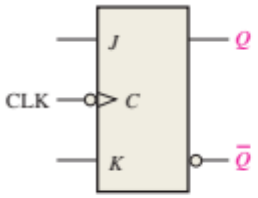
1. *Use A4 size paper for solution of each Question.*
2. *You are required to submit an assignment in hardcopy and also upload scanned copy on Google classroom.*
3. *Show all steps, otherwise marks will be deducted.*
4. *The deadline for submission is **23rd April, 2022.***
5. ***Copying is not allowed at all.** Any similarities among the submitted files of any student will result in **zero marks**.*

## CLO #0 (Total Marks -100)

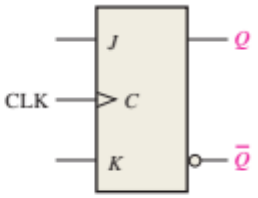
1. Two edge-triggered J-K flip-flops are shown in figure below, If the inputs are as shown, draw the Q output of each flip-flop relative to the clock, and explain the difference between the two. The flip-flops are initially RESET. [10]

In part a, the CLK is active low ∴ it gets triggered on negative edge & vice versa for part b.



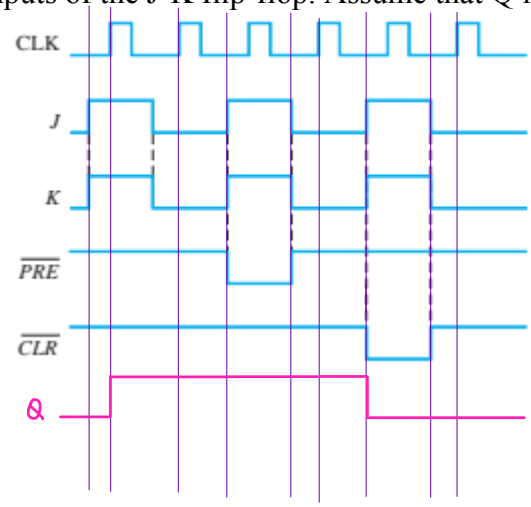



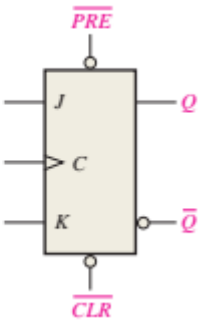

(a)



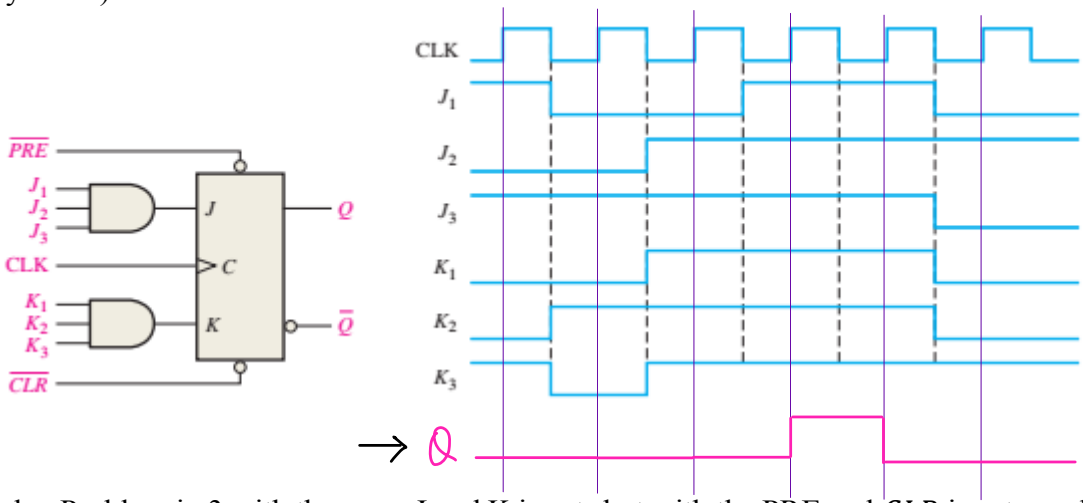

(b)

2. Determine the Q waveform relative to the clock if the signals shown in figure below, are applied to the inputs of the J-K flip-flop. Assume that Q is initially *LOW*. [10]

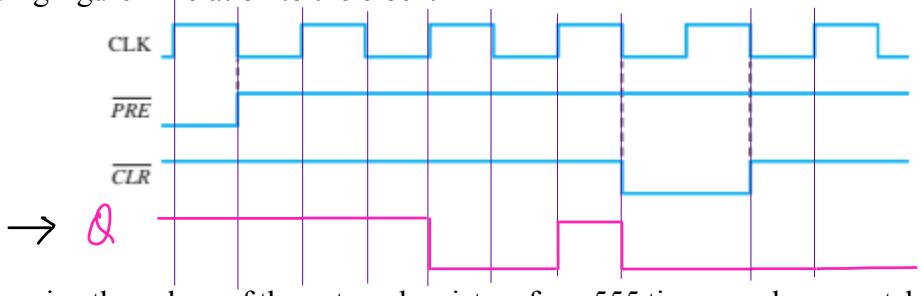


3. For the circuit in figure below, complete the timing diagram showing the Q output (which is initially $LOW$). Assume $\overline{PRE}$ and $\overline{CLR}$ remain $HIGH$. [10]




4. Solve Problem in 3 with the same J and K inputs but with the PRE and $CLR$ inputs as shown in following figure in relation to the clock. [10]

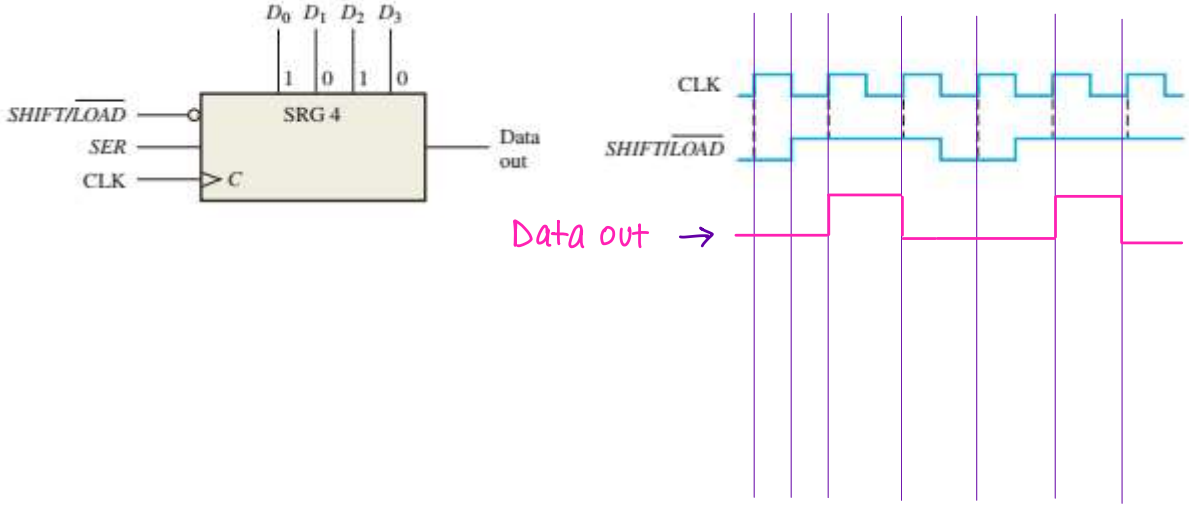



5. Determine the values of the external resistors for a 555 timer used as an astable multivibrator with an output frequency of 10 kHz, if the external capacitor C is 0.004 mF and the duty cycle is to be approximately 80%. [10]

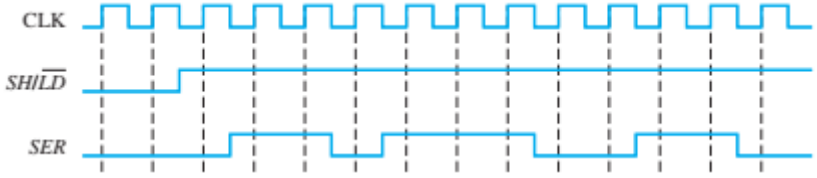
6. What is the state of the register in figure below after each clock pulse if it starts in the 101001111000 state? [10]




7. The shift register in figure below has $SHIFT/\overline{LOAD}$ and $CLK$ inputs as shown in figure below. The serial data input ($SER$) is a 0. The parallel data inputs are D0 = 1, D1 = 0, D2 = 1, and D3 = 0 as shown. Develop the data-output waveform in relation to the inputs. [10]



8. Develop the Q0 through Q7 outputs for a 8-bit Serial In Parallel Out shift register with the input waveforms shown in figure below. Here $A$ and $B$ are ANDed and used for Serial In to register. [10]

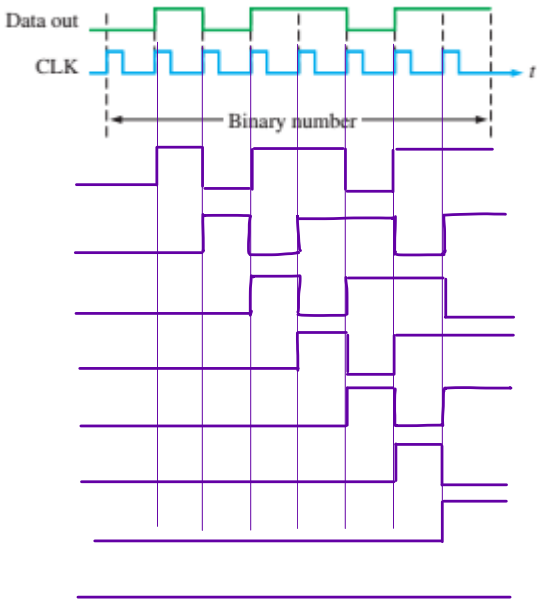



9. For the 8-bit bidirectional register in figure below, determine the state of the register after each clock pulse for the $RIGHT/\overline{LEFT}$ control waveform given. A $HIGH$ on this input enables a shift to the right, and a $LOW$ enables a shift to the left. Assume that the register is initially storing the decimal number seventy-six in binary, with the right-most position being the $LSB$. There is a $LOW$ on the data-input line. [10]




10. A leading-edge clocked serial in/serial out shift register has a data-output waveform as shown in figure below, what binary number is stored in the 8-bit register if the first data bit out (leftmost) is the $LSB$? [10]

0 1 0 1 1 0 1 1 ← LSB




1 1 0 1 1 0 1 0

$= (218)_{10}$

5. Determine the values of the external resistors for a 555 timer used as an astable multivibrator with an output frequency of 10 kHz, if the external capacitor C is 0.004 mF and the duty cycle is to be approximately 80%. [10]

$$T_H = 0.693(R_1 + R_2)C_1 \qquad T_L = 0.693R_2C_1$$

$$T = T_H + T_L = 0.693(R_1 + 2R_2)C_1$$

$$f = \frac{1}{T} = \frac{1}{0.693(R_1 + 2R_2)C_1}$$

$$10 \times 10^3 = \frac{1}{0.693(R_1 + 2R_2) \times 0.004 \times 10^{-3}}$$

$$R_1 + 2R_2 = 36.075 \quad \text{— (1)}$$

$$t_W \Rightarrow \frac{t_H}{T} \times 100 \Rightarrow \frac{R_1 + R_2}{R_1 + 2R_2} \times 100$$

$$0.8 = \frac{R_1 + R_2}{R_1 + 2R_2} \quad \text{— (2)}$$

$$R_1 = 36.075 - 2R_2$$

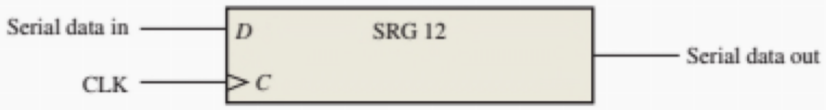
$$0.8 = \frac{36.075 - R_2}{36.075}$$

$$R_1 = 21.645 \qquad R_2 = 7.215$$

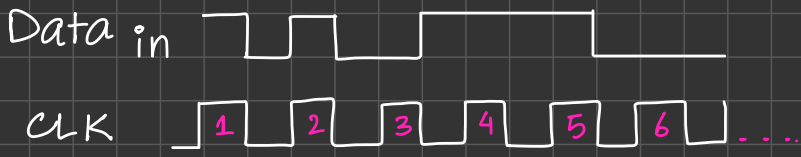
SH/$\overline{LD}$ is low → asynchronous

SH/$\overline{LD}$ is high → synchronous (rising edge)

6. What is the state of the register in figure below after each clock pulse if it starts in the 101001111000 state? [10]




$D_0 \ldots D_{11}$ = 101001111000




clock pulse 1 = 0

clock pulse 2 = 0

clock pulse 3 = 0

clock pulse 4 = 1

clock pulse 5 = 1

clock pulse 6 = 1

clock pulse 7 = 1

clock pulse 8 = 0

clock pulse 9 = 0

clock pulse 10 = 1

clock pulse 11 = 0

clock pulse 12 = 1

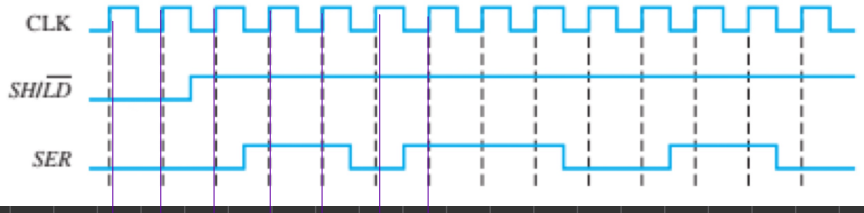
$D_0 \ldots D_{11}$ = 000111100101

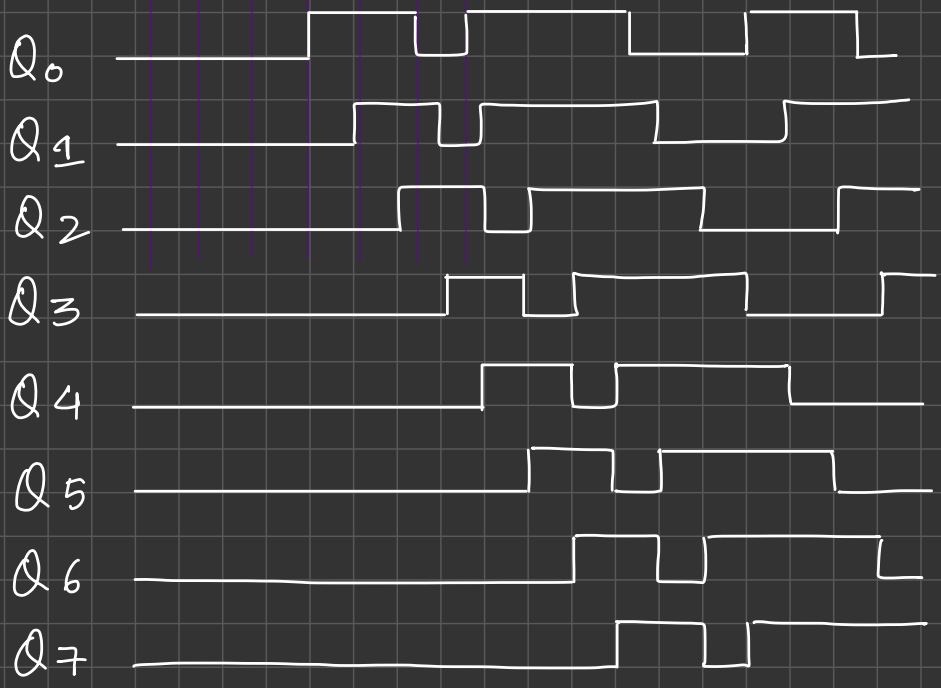
output

↑ first

↑ last

8. Develop the Q0 through Q7 outputs for a 8-bit Serial In Parallel Out shift register with the input waveforms shown in figure below. Here *A* and *B* are ANDed and used for Serial In to register. [10]

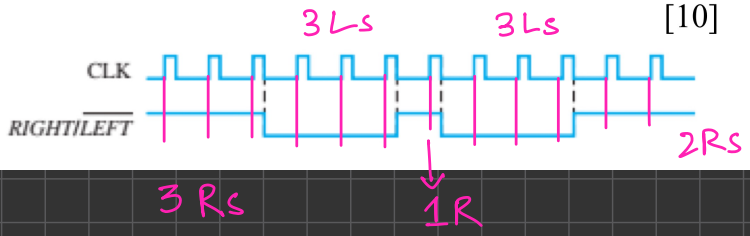



Each time Q is shifted by 1 pulse.

9. For the 8-bit bidirectional register in figure below, determine the state of the register after each clock pulse for the $RIGHT/\overline{LEFT}$ control waveform given. A *HIGH* on this input enables a shift to the right, and a *LOW* enables a shift to the left. Assume that the register is initially storing the decimal number seventy-six in binary, with the right-most position being the *LSB*. There is a *LOW* on the data-input line. [10]




76 = 01001100

(R) First 3 pulse : 00001001

(L) Next 3 pulse : 01001000

(R) Next 1 pulse : 00100100

(L) Next 3 pulse : 00100000

(R) Next 2 pulse : 00001000